

DarwinSync AI — Autonomous Migration Agent

FORWARD DEPLOYED ENGINEER (FDE) ARCHITECTURE DEFENSE

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1. GUIDING PRINCIPLE: PRIVACY-FIRST & THE ESCALATION BOUNDARY

“Automate transformations that are reversible, verifiable by code, or low-blast-radius; escalate decisions that are irreversible, high-blast-radius, or genuinely ambiguous.”

Coming from an AI Systems and secure multi-tenant infrastructure background, my baseline rule is that **client PII and sensitive compensation data must never leave the client perimeter to third-party LLMs**. Even with local models, AI is strictly an advisory assistant — it can suggest statistically optimal matches, but business-critical decisions depend on organizational context only a human domain expert knows.

SITUATION / DATA PATTERN	AGENT ACTION	ESCALATE?	ARCHITECTURAL RATIONALE
Whitespace, Case, Phone (+91)	Auto-normalizes to E.164	NO	100% reversible, regex-verifiable, zero data loss.
Date Format (Any day > 12)	Normalizes entire column	NO	Single row (e.g. 24/03/2021) provides mathematical proof of DD/MM.
Date Format (All days ≤ 12)	Holds format for confirmation	YES	04/05/2021 is ambiguous (Apr 5 vs May 4). Guessing permanently corrupts hire dates.
High-Confidence Header Match	Auto-maps to Darwinbox	NO	Statistically distinct score (≥ 0.75, margin ≥ 0.20) across aliases & type profiling.
Ambiguous Column (ref)	Surfaces top-2 with LLM context	YES	Tie between employee_id vs manager_id. Wrong choice mangles org hierarchy.
Sensitive Fields (Salary, DOB)	Batched 1-click confirmation	YES	Load-bearing fields require human sign-off once; remembered permanently.
Exact Multi-File Duplicate	Auto-merges & fills null gaps	NO	Corroborating identity; non-conflicting fields enrich employee record safely.
Near-Duplicate (Planted Case 4)	Flags high name match + same DOB	YES	Merging two distinct people with differing IDs into one Darwinbox record is irreversible.
Salary Conflict (HRIS vs Payroll)	Withholds auto-merge	YES	Payroll discrepancy (£80k vs £88k) cannot be auto-picked without legal liability.
Push 503 Transient Error	Exponential backoff & jitter	NO	Network/service blip; succeeds silently on retry without polluting review queue.
Target System Outage (3x 5xx)	Circuit breaker + Batch Rollback	YES	Issues compensating DELETE calls for active batch to avoid partial dirty state.

2. DELTA ENGINEERING: RESOLUTION MEMORY

The differentiator of an FDE is not calling an LLM API; it is the **operational delta** wrapped around it. Asking consultants to re-review identical ambiguities on every recurring migration dump causes reviewer fatigue.

When a human resolves an item (e.g. mapping ref → employee_id or LOA → On Leave) and checks *Remember as rule*, the decision is fingerprinted into the client-scoped rules table. On subsequent runs, the agent checks resolution memory *first*.

7 → 3
ESCALATIONS OPENED

-57%
REVIEW EFFORT

88.5% → 95.1%
AUTONOMY SCORE

3. PRODUCTION ROADMAP: WHAT I'D BUILD NEXT

- Topological Entity Push:** Sequence push ordering (Departments → Cost Centers → Managers → Employees) to prevent foreign-key rejection.
- Cross-Tenant Synonym Embeddings:** Cluster anonymized header embeddings across clients to suggest pre-warmed synonym mappings while preserving strict tenant isolation.
- Automated Schema Drift Alerts:** Monitor recurring client SFTP drops; detect newly added columns or shifted date conventions and auto-propose draft migration diffs.
- Compensating Saga Ledger:** Upgrade batch rollbacks to distributed sagas for multi-entity transactional purity.